

Implementing Configuration Management



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Module Overview



Reviewing the application code

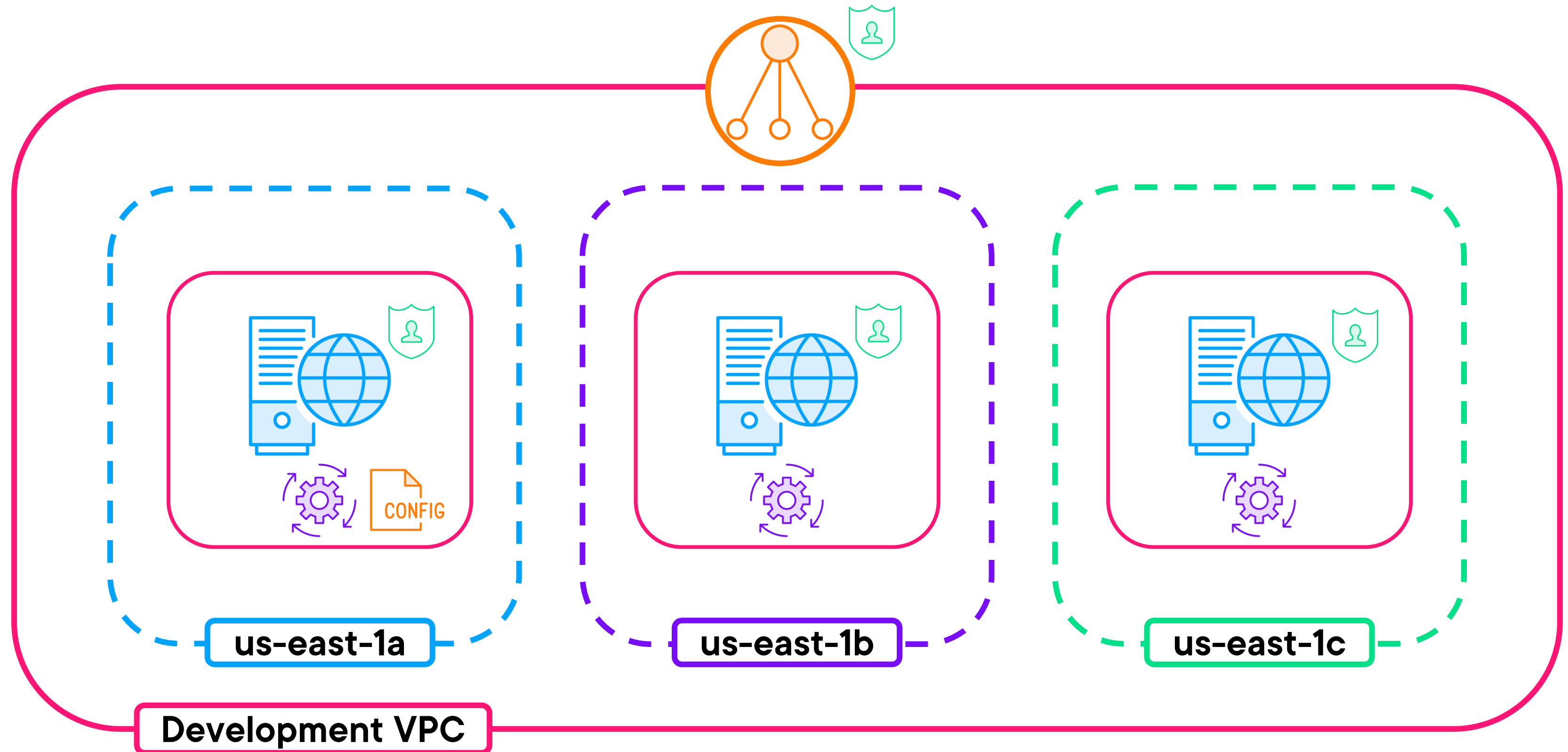
Configuration management overview

Replace provisioners with Ansible

Replace null_resource with terraform_data



Taco Wagon Architecture



Recommendations



Remove input variables for network info

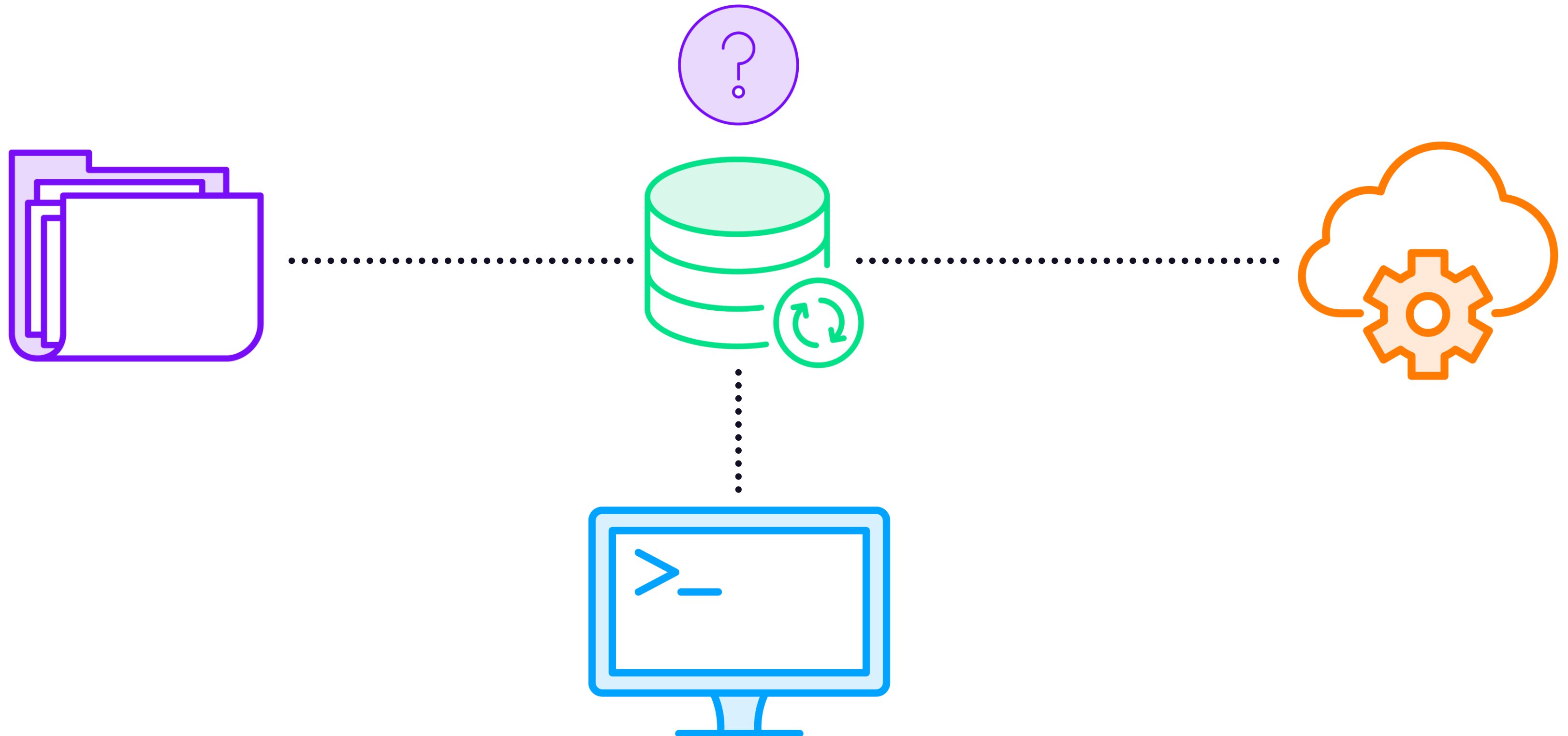
Use Ansible to deploy application

Switch to the terraform_data resource

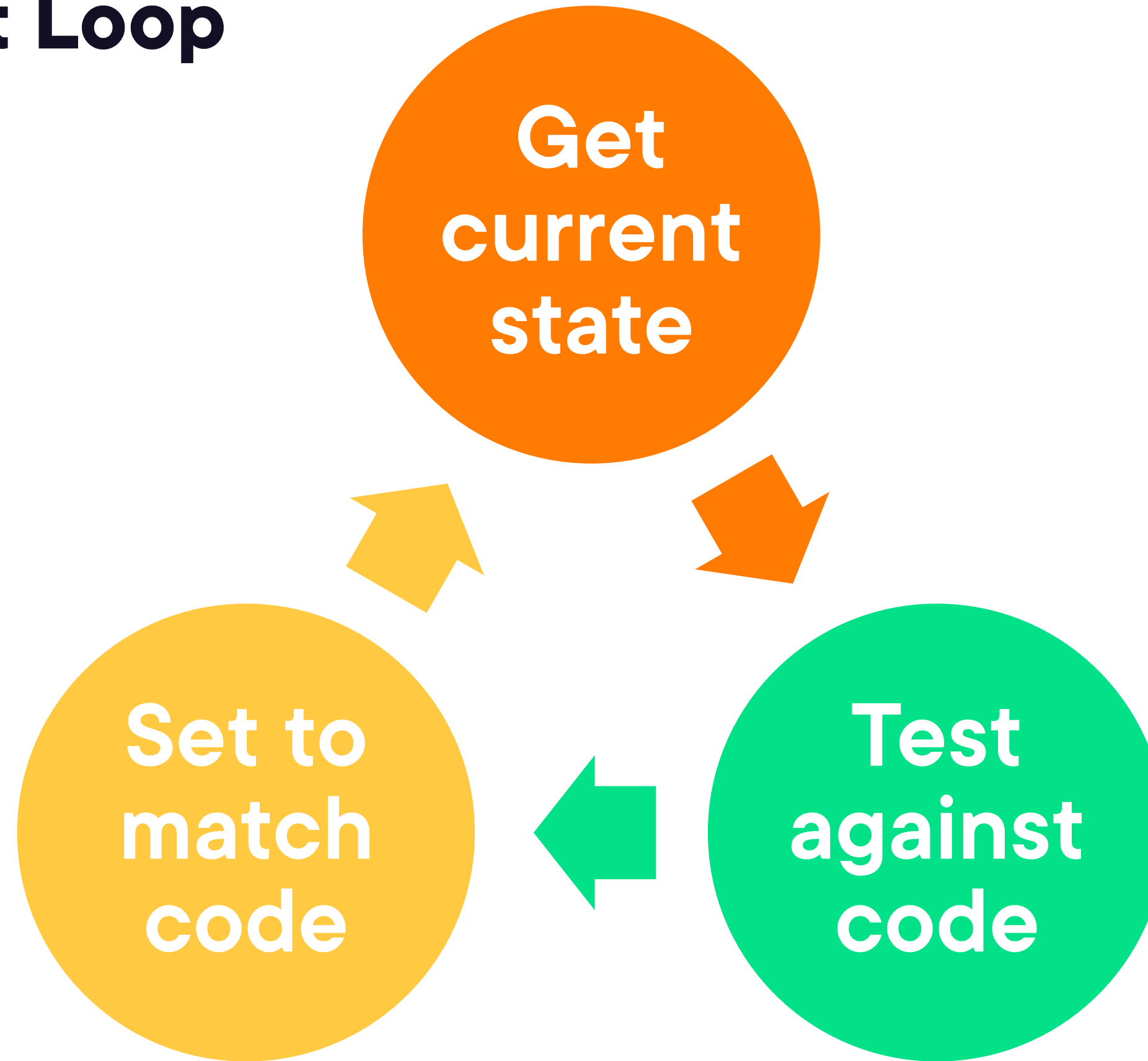
Store the API key securely



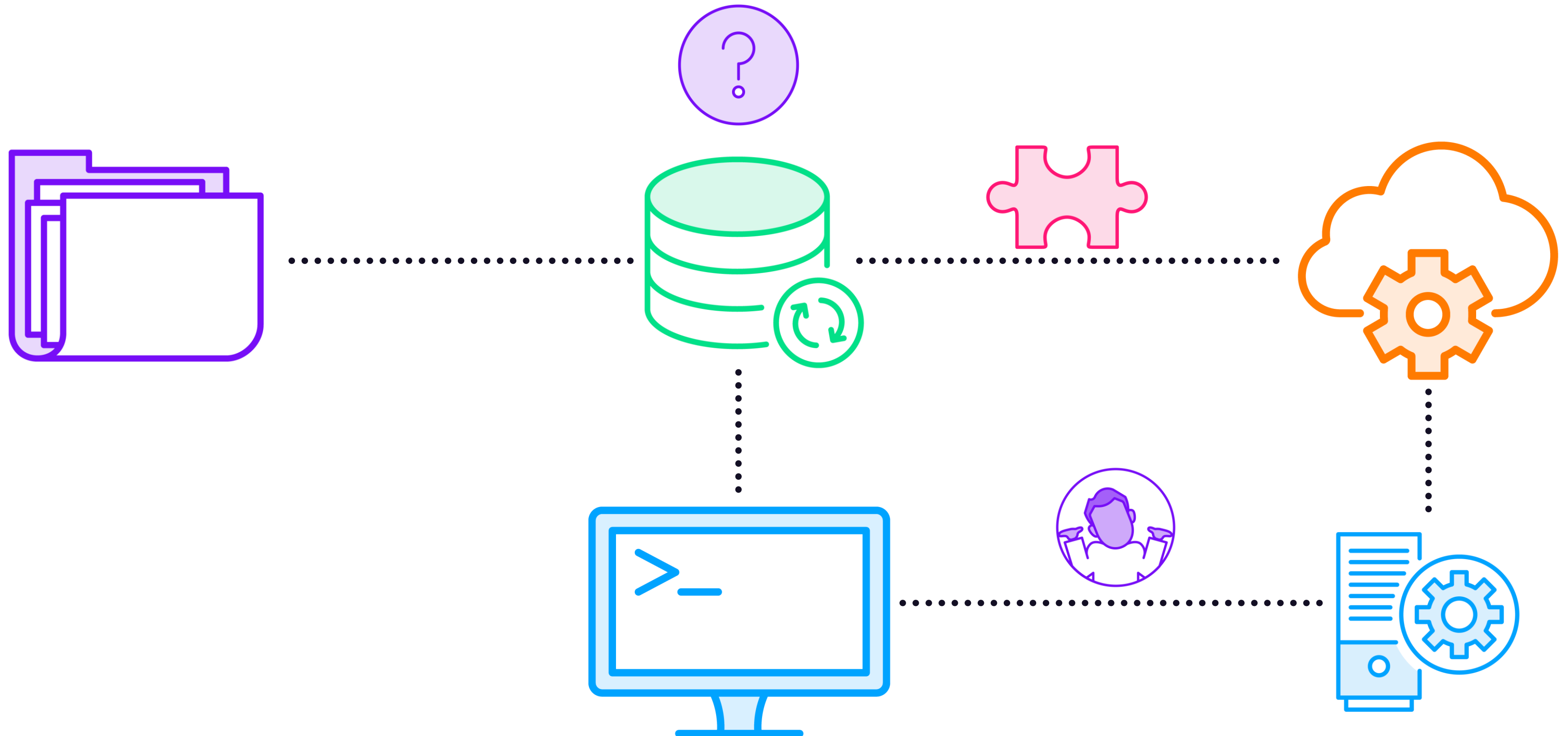
Terraform Limitations



Get Test Set Loop



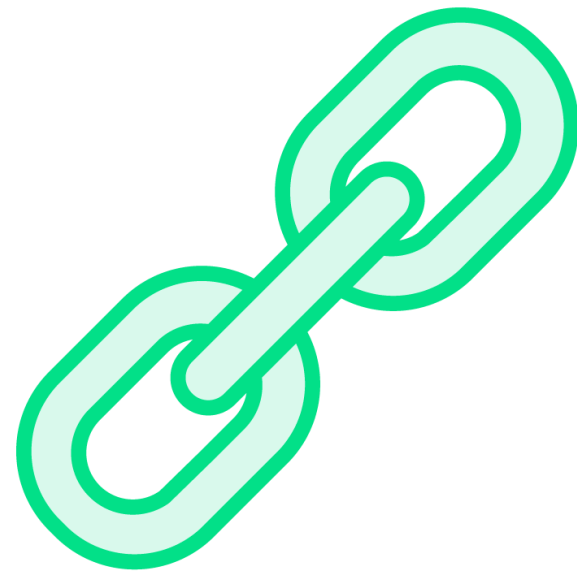
Terraform Limitations



Configuration Management Concepts



Identification
What to manage



Control
How to manage



Accounting
How to report

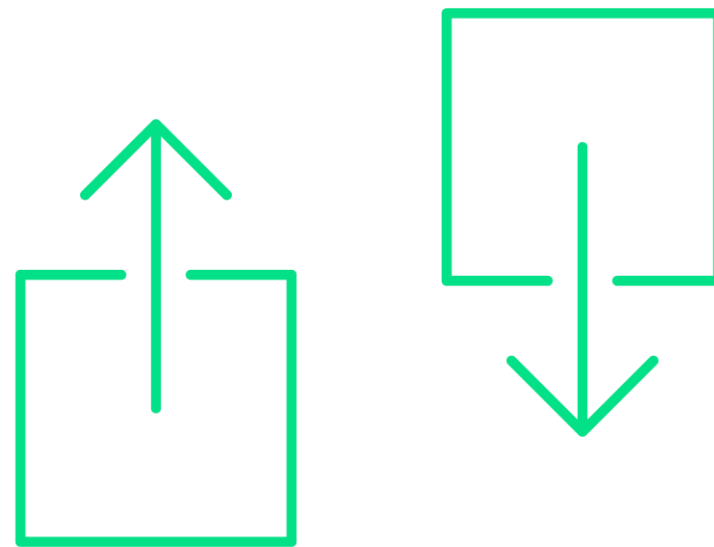


Verification
How to validate

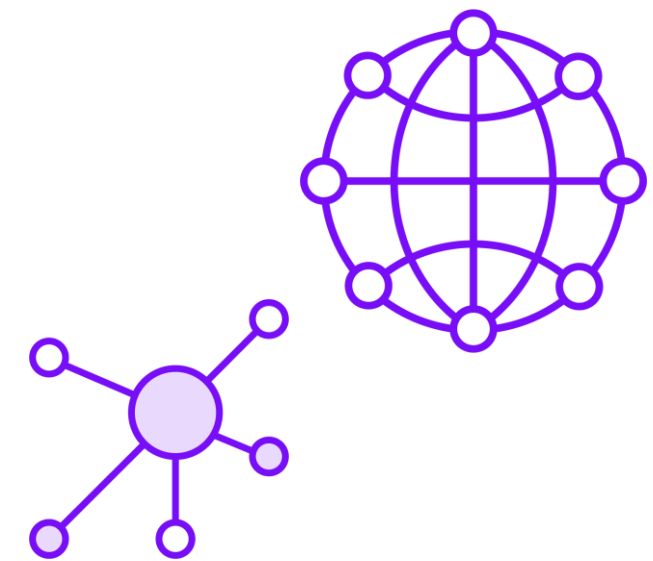
Deployment Options



Terraform hand-off

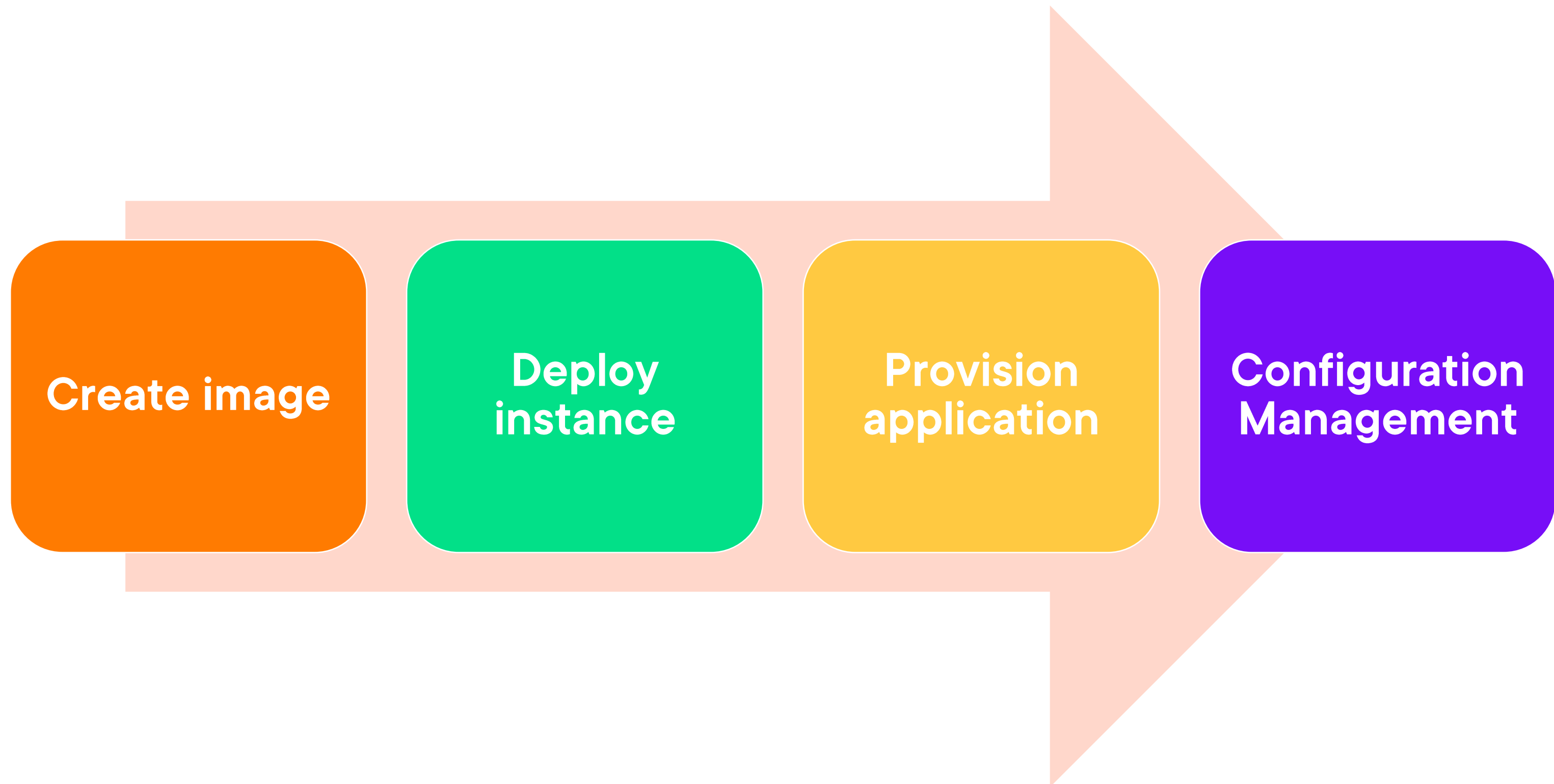


Push or pull

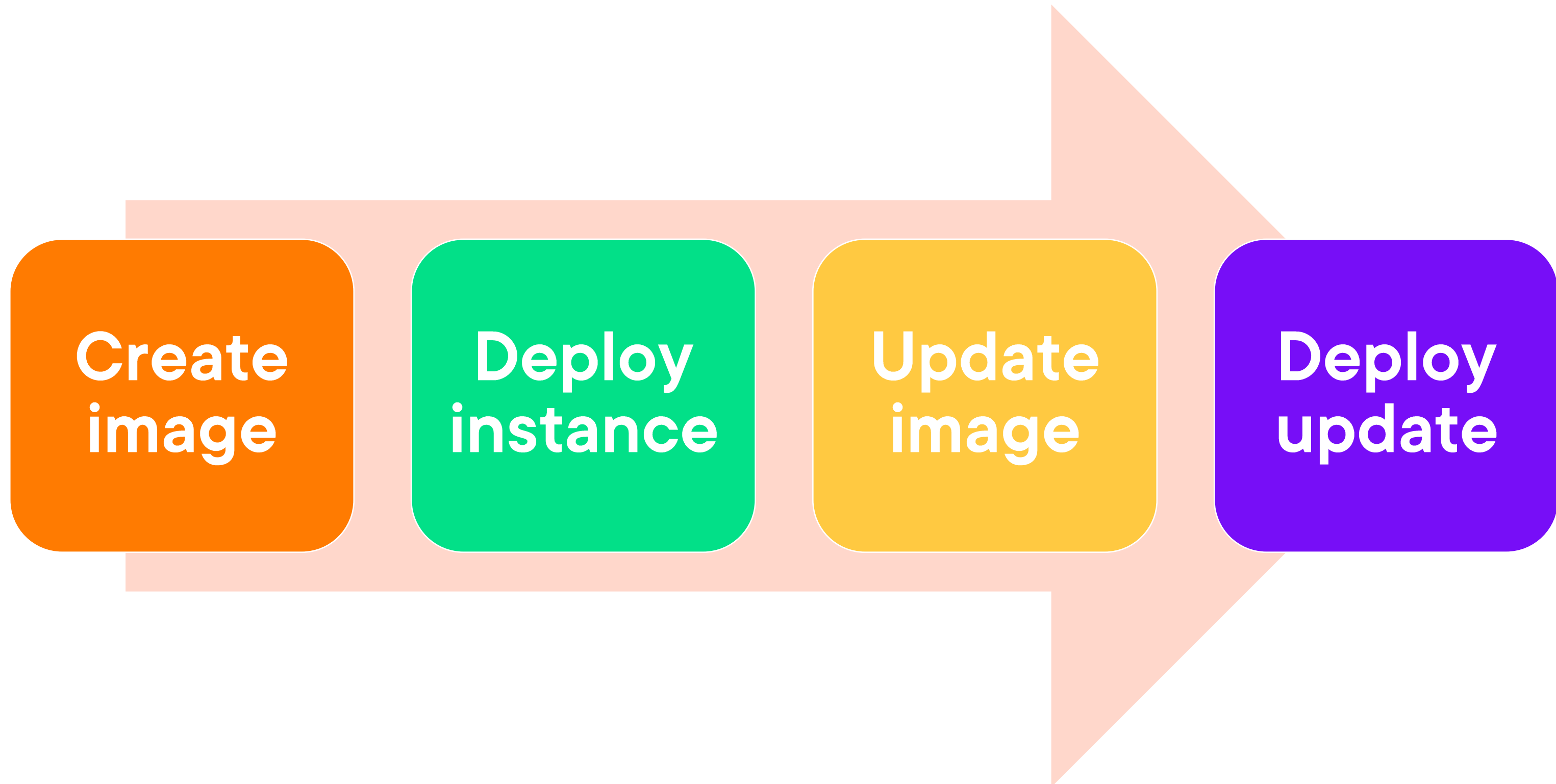


**Centralized or
distributed**

Traditional Deployment



Immutable Deployment



Ansible Overview



Open-source configuration management tool

Agentless operation

Connects remotely or runs locally

Playbooks in YAML

- Tasks and modules

Inventory file in YAML

Centralized control

- Red Hat Ansible Automation
- Ansible Semaphore



Demo



Pull playbook from GitHub repo

Remove some provisioners

Add userdata argument to EC2 instances



null_resource

datasources.tf

```
resource "null_resource" "example" {
  triggers = {
    # Watch list of instance IDs
    instance_ids = join(",", aws_instance.main.*.id)
  }

  provisioner "remote-exec" {...}
}
```



terraform_data

datasources.tf

```
resource "terraform_data" "example" {
  triggers_replace = [
    # Watch list of instance IDs
    join(",", aws_instance.main.*.id)
  ]

  provisioner "remote-exec" {...}

  # Force an update of an IAM policy
  input = var.iam_version_update
}

resource "aws_iam_policy" "example" {
  lifecycle {
    replace_triggered_by = [terraform_data.example]
  }
}
```



Module Summary



**Configuration management tools or
Terraform**

**Replace provisioners with configuration
management**

**Replace the `null_resource` with
`terraform_data`**



Up Next:

Dealing with Sensitive Data

